



Shahriar Khan

Software Engineer

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PROFESSIONAL SUMMARY

CSE graduate (CGPA 3.92/4.00) with experience building full-stack web and mobile products, backend APIs, computer-vision pipelines, and SaaS platforms. Combines React/Next.js and Flutter development with Python-based ML and deployment. Projects recognized by the Dhaka Metropolitan Police. 1.5 years of teaching experience as an Undergraduate Teaching Assistant. Open to full-time roles and internships in software engineering, mobile development, and AI/ML engineering.

TECHNICAL SKILLS

Languages: Python, C++, Java, TypeScript, JavaScript, Dart, SQL

Web: Next.js, React, Node.js, Express, Flask, Tailwind CSS, REST APIs, Socket.IO

Mobile: Flutter, React Native, Expo, Riverpod, GoRouter, Firebase, Supabase

AI / Vision: Python, PyTorch, TensorFlow, YOLOv11, scikit-learn, PySpark MLlib, OpenCV, Grad-CAM, LIME

Data / DevOps: PostgreSQL, MongoDB, SQLite, Docker, Git, GitHub Actions, Vercel, DigitalOcean, Linux

Problem Solving: 100+ Codeforces problems solved | Strong proficiency in Data Structures & Algorithms

EXPERIENCE

East West University

November 2024 – April 2026

Undergraduate Teaching Assistant, Department of CSE

Dhaka, Bangladesh

- Supported programming labs, reviewed code, evaluated assignments, prepared academic resources, and helped students debug Python, C++, and Java projects over three semesters.

SELECTED PROJECTS

TrafficGuard AI – Traffic Surveillance & CCTV Analytics

[GitHub](#)

Tools: Python, Flask, Flutter, YOLOv11, OpenCV, RTSP, Gemini AI

- Built an end-to-end multi-camera pipeline detecting 16 vehicle classes with speed estimation, wrong-way analysis, crowd detection, adaptive signal control, and a Gemini AI assistant.
- Trained and compared six YOLO architectures with ablation studies on a custom CCTV dataset. Received official recognition from DMP, Spectrum Engineering, and Cybernetics Hi-Tech Solutions.

NeoFocus – Gamified Productivity OS

[GitHub](#)

Tools: Flutter, Dart, Supabase, Riverpod, GoRouter

- Built a full-featured mobile productivity app with Pomodoro timer, XP/level system, habit tracker, task management, and Supabase-powered cloud sync and authentication.

JetStream – Hybrid Music Streaming Platform

[GitHub](#)

Tools: React, TypeScript, Node.js, Expo, React Native, WebAudio API

- Developed a premium music platform spanning a glassmorphic React web UI, cross-platform Expo mobile app, and a Node.js audio extraction server with Deezer metadata integration.

Banana XAI – Explainable Image Classification

[Live](#) | [GitHub](#)

Tools: PyTorch, Streamlit, Grad-CAM, Grad-CAM++, Eigen-CAM, LIME

- Benchmarked 12 CNN and vision-transformer models with four explainability methods in an interactive comparison dashboard.

PromptMaster Pro – AI Learning Platform

[Live](#) | [GitHub](#)

Tools: React, GSAP, Supabase, Lemon Squeezy

- Launched a responsive SaaS platform serving 200+ members and 59+ lessons, with authentication, Stripe/Lemon Squeezy payments, and animated product pages.

NutriScan AI – Food Recognition & Nutrition Tracker

[GitHub](#)

Tools: Flutter, Dart, Firebase, Google Gemini AI

- Built an AI-powered mobile app that identifies food items from photos using Gemini AI vision and provides nutritional breakdowns and calorie tracking, with Firebase authentication and cloud storage.

Student Expense Tracker – Finance App with OCR

[GitHub](#)

Tools: Flutter, Dart, Firebase, SQLite, OCR

- Developed a feature-rich Flutter expense tracker with OCR receipt scanning, budget alerts, savings goals, smart notifications, and PDF export, backed by Firebase Auth and local SQLite storage.

Election Analytics 2026 – ML Prediction Dashboard

[Live](#) | [GitHub](#)

Tools: Next.js, TypeScript, PySpark MLlib, React Native

- Built a predictive dashboard achieving 84.78% classification accuracy, with Bangladesh choropleth and parliamentary-seat visualizations, plus a React Native mobile companion.

EnergyIQ – IoT Energy Monitoring Dashboard

[Live](#) | [GitHub](#)

Tools: Next.js, Framer Motion, TOMZN Wi-Fi Smart Meter, GitHub Pages

- Connected real smart-meter hardware to a live monitoring dashboard with tariff-based cost projections, scroll-driven animated storytelling, and automated deployment to GitHub Pages.

EDUCATION

B.Sc. in Computer Science & Engineering, East West University, Dhaka (2022–2026)

CGPA: 3.92/4.00 | Dean's List

Higher Secondary Certificate, Government Science College, GPA: 5.00 (2020)

Secondary School Certificate, National Ideal School, GPA: 5.00 (2018)

TEST SCORES & LANGUAGES

TOEFL iBT: 93 (January 2024) | **Bengali**: Native | **English**: Professional working proficiency

PUBLICATIONS

- Fatiha Mahbub Mim, Mossammat Ishrat, Md Tanvir Chowdhury, Md Tanzid Mollah, **Shahriar Khan**, K. M. Safin Kamal, and Ahmed Wasif Reza. “[Dust Cloud Detection: A Machine Learning Approach](#).” In *Data Mining and Information Security*, LNNS 1915, pp. 77–86, Springer, 2026.
- Md Tanzid Mollah, Md Tanvir Chowdhury, Ahamed Jaman Jehad, **Shahriar Khan**, and Md Fokrul Akon. “[Data Transmission in IoT Networks with Minimal Energy Consumption](#).” *IJERT*, vol. 14, no. 05, IIRA 5.0, 2026.

RECOGNITION & SCHOLARSHIPS

Project Recognition Letters from Dhaka Metropolitan Police (DMP), Spectrum Engineering Consortium, and Cybernetics Hi-Tech Solutions (2026)

Dean's Merit Scholarship (2025) | **Merit Scholarship & Dean's List** (2024) | **Full Tuition-Free Merit Scholarship** (2022)